

Use Case Manual Payment Authentication



A Use Case Manual is a description of a specific use case for the EU Digital Identity Wallet. It includes an introduction to the use case and the user journey, the state of play of the use case development, as well as references to the relevant technical specifications and infrastructures, legislation, and governance structures.

The Use Case Manual is designed for the Wallet implementers, Member States and other stakeholders interested in the development of the Wallet and its use cases.

The Use Case Manual is not intended to contain full detailed technical specifications for the use case. However, where such specifications are available, the Use Case Manual contains pointers to them.

Description

The Payment Authentication use case enables the authentication of payment requests through EU Digital Identity Wallets (EUDI Wallets) for both online and in-store transactions. It can leverage existing payment infrastructures such as SEPA Credit Transfer (SCT), including instant and traditional account-to-account payments, and card payments, complementing current card-based and A2A systems without causing disruption. For online transactions, this integration model supports enriched checkouts where identity attributes (e.g. age) or attestations (e.g. driving license) can be securely shared as part of the payment flow, enabling use cases such as age verification when paying online for an age-restricted service, or presenting a driving license when paying for renting a car online.

With regard to authentication for electronic payments, EU Digital Identity users will be able to perform Strong Customer Authentication by using their EU Digital Identity Wallets, ensuring a high level of security and compliance with PSD2 (and with the future PSD3/PSR). In particular, the Wallet must enable the user to give his/her consent for a payment by authenticating transactional information (e.g., payment and payee information).

From a regulatory perspective, by the end of 2027, various service providers from different sectors (e.g., payment service providers in the financial sector) will be required to accept EU Digital Identity Wallets (EUDI Wallet), upon request of the user, where SCA is required by Union or national law or by contractual obligation (Article 5f of the EU Digital Identity Framework).

User Journey(s)

Prior to any payment transactions with the EUDI Wallet, the Payment Service Provider – PSP (e.g. a bank) – will offer its clients (Users) the option to link their Wallets as a means of payment authentication. This is a one-time registration process involving verification of both the User and the EUDI Wallet by the PSP, after which the PSP will issue dedicated payment-authentication credentials to the Wallet in the form of attestations.

After this, the User will be able to authenticate any subsequent payments with the Wallet by presenting these attestations.

[DISCLAIMER] The following preliminary (merchant-led) user journeys are only indicative and are subject to change based on updates to technical and legislative documentation, as well as outcomes of the Large Scale Pilots (LSPs). They should not be taken as definitive, and are not representing final configuration.

Scenario – Online Transaction:

1. User completes an online checkout and selects the payment method linked with his EUDI Wallet
2. Merchant sends a payment request to payment interface, including at least:

- Amount
 - Recipient
 - Request for verification of required attributes
3. Payment window on merchant website prompts user to scan QR code using the EUDI Wallet or to push a button to open the EUDI Wallet
 4. User opens the EUDI Wallet application
 5. EUDI Wallet displays:
 - Amount
 - Recipient
 - Request for verification of required attributesand EUDI Wallet asks for consent confirmation
 6. User reviews information and confirms by authenticating with biometric or PIN
 7. Wallet generates and signs a proof of payment authentication
 8. The Payment Service Provider receives the proof, verifies the user's attributes and executes the transaction
 9. Merchant receives a confirmation from the Payment Service Provider
 10. Payment window on merchant website closes. The user receives the good or service.

Scenario – In-Person Transaction:

1. At the checkout, merchant asks the User for payment authentication
2. Merchant generates a payment request and displays a QR code with their payment terminal
3. User scans the QR code using the EUDI Wallet application
4. EUDI Wallet app displays:
 - Merchant name, payment amount, and other payment related data
 - Required attributes to share
 - Additional information e.g. on security or data privacyand EUDI Wallet asks for consent confirmation
5. User reviews information and confirms by authenticating with biometric or PIN
6. Wallet generates and signs a proof of payment authentication
7. The Payment Service Provider receives the proof, verifies the user's attributes and executes the transaction
8. Transaction is confirmed and user receives goods

Use case impact

This use case enables secure, consent-based payment authentication directly through the EU Digital Identity Wallet.

Value for user/citizen

For citizens, the use of the EU Digital Identity Wallets to authenticate payments presents an improvement in security, privacy, and convenience. Users can authenticate a payment directly from their wallet with a biometric or PIN.

Selective disclosure: The user selects to share verified attributes (like age or address) alongside payment only when needed, without exposing full identity and remains in control of the personal data shared;

Easy to use: Pay via QR scan or by clicking a button and then store digital receipt, boarding pass or other credentials received from the merchant;

Combined presentations: one or more different types of attestations (e.g., age, driving license) from different authentic sources can be securely shared as part of a single payment flow;

Prevents fraud: Fraud attacks such as phishing and credential theft are harder to execute, as users never expose account details or passwords to merchants;

Increased trust in EU Digital Identity Wallet infrastructure: As payment authentication could be one of the most frequently used wallet interactions, it serves as a practical proof of reliability and security in everyday life;

Cross-service continuity: Use the same wallet to authenticate payments across banks, merchants, and platforms;

Privacy and consent: Authenticate payments without sharing static credentials or passwords.

Value for service providers

Service providers receive a standards-based, secure, and interoperable channel for SCA and consistent checkout experience across the single market. Service providers benefit from lower implementation costs and improved customer satisfaction.

Cost reduction: Ready-to-use new authentication solution via the Wallet that can reduce costs versus existing solutions;

Streamlined Compliance & Audit: The Wallet's built-in, certified security features streamline compliance with regulatory requirements (e.g., SCA under PSD2), reducing the need of external audits and compliance costs;

Increased Convenience for Merchants: Using government-verified age checks directly from the Wallet app helps merchants legally sell restricted items (like alcohol, tobacco, or gambling services).

This high-assurance verification significantly lowers the risk of fines and penalties for accidentally selling to minors;

Risk Mitigation for Banks/PSPs: Benefit from the reduced risk associated with fraud and unauthorised transactions due to the Wallet providing a higher, standardized assurance level for the identity of the payer and the security of the authentication.

Value for public authorities

Use of the EU Digital Identity Wallets in payment authentication supports financial inclusion, fraud mitigation, and harmonised compliance across Member States. Regulators benefit from improved auditability and increased trust in the identity of payment initiators. The wallet's ability to carry cryptographically verifiable identity and consent signals enhances enforcement of AML/CFT (Anti-Money Laundering and Combating the Financing of Terrorism) controls without increasing complexity for users.

Regulatory harmonisation: Aligns payment authentication with the eIDAS (electronic Identification, Authentication and trust Services) trust framework, reducing fragmentation across national implementations;

Auditability: Wallet-backed payment authentication creates secured proof of transaction approval;

Increased Trust in Digital Transactions: Provides regulators and public bodies with increased trust in the identity of payment initiators and service providers, reducing risk within the digital economy;

Public Service Modernisation: Shows progress in digital transformation of public and private sector services by providing modern digital payment services based on privacy preservation and cross-border interoperability.

Economic value and potential market size

Users

The potential user base includes all EU residents with a bank account or payment method. According to Eurostat, over 86% of EU adults have made an online purchase in the past year.

Issuers

The primary issuers of credentials are banks and payment service providers (PSPs) who integrate the EUDI Wallet into their SCA infrastructure. For financial institutions, wallet integration offers both cost reduction and value-added service potential. While the EU Digital Identity Wallets themselves are not monetised, banks can differentiate themselves by offering frictionless SCA and seamless identity–payment experiences.

Relying Parties

All enterprises that accept digital payments can play the role of Relying Parties when a user authorises a payment (both online and in-store) through the EU Digital Identity Wallet. Considering online transactions, Relying Parties would be represented by enterprises that operate in e-commerce. As per European Commission's [E-commerce statistics](#) report, the share of enterprises that had e-sales in 2023 resulted in 23.8%. [Eurostat reported](#) that in 2023 there were about 33 million enterprises in EU (27 MS countries), meaning that almost 8 million enterprises did business online.

Transactions

In 2023, based on ECB data, the euro area recorded approximately 140 billion non-cash payment

transactions, including card payments, credit transfers, and direct debits. In 2024 this number increased to 150 billion. [Source 1](#) [Source 2](#)

In terms of payment transaction value, the euro area accounted for €230.4 trillion in 2024, compared to €225.3 in 2023. [Source](#)

The [Study on the payment attitudes of consumers in the euro area](#) from 2024 (ECB) observes rise of preference for card and other cashless payments at POS, as well as rising preference for online payments in relation to fall in share of POS payments, whose value dropped from 76% in 2019 to 58% in 2024. In the same period, the share of online day-to day payments increased from 18% to 36% in value.

The most frequently used instrument for online payments in 2024 was cards, representing 48% of number of transactions (vs. 51% in 2022). However, the share of e-payment solutions, i.e. payment wallets and mobile apps, rose from 26% in 2022 to 29% in 2024. These include services like PayPal, big tech wallets (Apple Pay, Google Pay/Wallet) and other mobile apps (e.g., iDEAL in the Netherlands).

Remote / online payment flows are disproportionately exposed to fraud, compared to in-person (POS) or non-remote flows. As per 2024 Report on Payment Fraud by ECB, the payment fraud reported by the industry across the European Economic Area amounted to €4.3 billion in the year 2022 and €2.0 billion in the first half of 2023. Most payment fraud in value terms was related to credit transfers and card payments, across all three reference periods analysed. [Source](#)

Synergies with other use cases

Payment Authentication can act as a foundational component across numerous scenarios of digital service delivery and therefore creates synergies with the majority of EUDI Wallet use cases. For example, the following use cases show strong synergies with Payment Authentication:

- *eSignature:*
In many financial or contractual processes, both payment and signature must occur for transaction to execute, for example, completing an online service contract that requires both a signature and a fee. The EU Digital Identity Wallets can handle both operations.
- *Age Verification:*
Wallet-based payment flows combine age or eligibility verification with transaction authorisation for regulated products or services. When a user pays for age-restricted goods, the same wallet flow can deliver both the payment proof and a selectively disclosed age attestation. No need for merchants to perform separate age checks or store sensitive personal data.
- *Natural or legal person representation:*
This synergy ensures legal certainty in delegated payments.

Use case implementation

Existing implementations

Several EU Digital identity Wallet Large-Scale Pilots (LSPs) have successfully tested the use of the EU Digital Identity Wallets in payment scenarios. The EWC (European Wallet Consortium) and the NOBID (Nordic-Baltic eID) consortium successfully piloted SCA specifications, demonstrating their application in payment transactions and assessing related regulatory and user experience aspects. Overall, this resulted in five payment use cases of various scenarios, from in-person to online payment authentication with age verification, as well as exploring the possibility of payment initiation. Together, they have converged on a joint solution that aims to introduce a harmonised, EU-wide model for integrating SCA and attribute sharing, ensuring the integration of EU Digital Identity Wallets with card-based and account-based payments while minimising disruption to existing payment infrastructures. Building on the outcomes of the pilots, ongoing initiatives such as WeBuild and Aptitude continue testing and refining the integration of the EU Digital Identity Wallets in payment scenarios.

In particular, EWC Payment Taskforce main focus was on piloting and designing the use of the European Digital Identity Wallet (EUDI Wallet) as a means of Strong Customer Authentication (SCA) for online payments, aligning with the eIDAS2 requirements for banks to accept EUDI Wallets for SCA. Beyond SCA requirements, EWC's wide piloting scope included also payment initiation, transforming EUDI Wallets into true payment wallets holding payment credentials. They emphasized that combining the payment and identity credentials enabled "Fast Checkout", which 80% of surveyed users found to be easier than standard checkout.

In the end, EWC piloted two flows in production stage. Firstly a "bank-led"¹ authentication flow, where the payer's bank interacts with the EUDI Wallet to authenticate the payer in a €25 ferry ticket purchase. And secondly, a "merchant-captured"² authentication flow, where the merchant interacts with the EUDI Wallet to capture the authentication data before providing it back to the payer's bank for the approval. Users were able to use Fast Checkout with the EUDI Wallet to authenticate the payment, consent to share PhotoID, student attestation and loyalty card attestation. When the payment concluded, payment receipt and boarding pass attestations were both issued in the EUDI Wallet without further interaction from the user.

[Demo – Online payments with an EUDI Wallet](#)

[Demo – Fast checkout with an EU Digital Identity Wallet](#)

NOBID consortium was made up of a group of Nordic and Baltic countries (Iceland, Norway, Denmark, Latvia), together with Italy and Germany, and focused on domestic and cross-border payments use case, aiming to create a seamless, secure and inclusive payment ecosystem across Europe for citizens and businesses. The pilot contributed to defining requirements and technical specifications, and entailed robust implementations and national usability testing in all six partner countries. This ultimately validated key architectural components of the EU Digital Identity framework and pointed to practical interoperability across Member States. NOBID concluded in September 2025 and its efforts resulted in three relevant payment demos, the in-person payment and age verification, the online payment and age verification, and the full payment flow. An additional demo has also been developed in November 2025 showcasing the in-person payment with age verification and digital receipt.

[Demo – In-person payment and age verification](#)

[Demo – Online payment and age verification](#)

[Demo – Full payment flow](#)

[Demo – In-person payment and age verification](#) and digital receipt.

WE BUILD is an ongoing large-scale, European collaboration aiming to design, implement, and test a trusted digital business identity ecosystem based on the European Digital Identity Wallet (EUDIW) framework. Out of 13 projected use cases, 4 are related to payments, including Consumer Banking and Consumer payments & Business Banking and Payments, with VISA Europe as domain lead.

APTITUDE has one of the objectives dedicated to enhancing the wallet to support banking and payment features, not only through technical implementations like protocols, security, and cryptography but also by ensuring an excellent user experience necessary for the wallet's adoption in financial use-cases. To achieve this, APTITUDE, with DSGVO as domain lead, will release the Payment and Banking EUDIW Extension (PBEE) Framework, standardising payment features across wallet implementations. This framework will enable wallet providers to integrate payment functionalities seamlessly, like mobile SDKs (Software Development Kit) used in banking apps.

Legislation

[Commission Delegated Regulation \(EU\) 2018/389](#) (RTS on SCA and secure communication under PSD2) – lays down the requirements for SCA, which needs to be complied with when accessing a payment account online and for initiating electronic payments, or carrying out any action through a remote channel which may imply a risk of payment fraud or other abuses.

[Users should be able to authenticate themselves and their electronic payments](#) securely and conveniently using their Wallet Units. A Wallet Unit facilitates complying with strong customer authentication (SCA) requirements for electronic payments, ensuring a high level of security and compliance with [Article 97](#) of the PSD2 (and with the future PSD3/PSR).

[Regulation \(EU\) 2024/886](#) on Instant Payments – requires EU banks to offer SCT Inst payments by default and mandates faster, irrevocable execution.

[Regulation \(EU\) 2024/1183](#) on the European Digital Identity Framework (and related Commission Implementing Regulations) – enables the creation of a universal, trustworthy, and secure European Digital Identity Wallet.

[Regulation \(EU\) 2014/910](#) provides regulatory environment on electronic identification and trust services for electronic transactions in the internal market.

¹Also known as issuer-requested flow in technical specifications (ts-12)

²Also known as Third-party-requested flow in technical specifications (ts-12)

Technical specifications, software libraries and infrastructure

Disclaimer: the technical specifications will be determined in upcoming implementing Acts and are subject to change.

Developers should comply with [Specification of Electronic Payments Strong Customer Authentication \(SCA\) Implementation with the Wallet](#) document which defines the behaviour of the Wallet Units when receiving a presentation request of attributes from a SUA Attestation used for SCA in payments (called “SCA Attestation”). The current version of the document focuses on [SD-JWT-VC] format and [OID4VP] presentation protocol only.

Strong Customer Authentication (SCA) for payments is mandated under PSD2 Article 97(1)(b) for electronic payment initiation, as well as accessing a payment account or other User’s actions that present a risk of fraud or other abuses. By design, the EUDI Wallet supports all three authentication factors required by PSD2 (knowledge, possession, inherence) to enable SCA. More details on the use of the Wallet for SCA purposes can be found in the [Architecture Reference Framework \(ARF\)](#) latest release (2.7.3 in November 2025), under [section 2.6.4](#).

The Payment Authentication use case is designed to work within the existing payment infrastructure in Europe, enabling integration with both card-based and account-to-account (A2A) systems. It operates in VC mode using OpenID for Verifiable Credential Issuance (OID4VCI) and OpenID for Verifiable Presentations (OID4VP) protocols. Wallet-based User authentication can be triggered in response to a payment request initiated by a merchant, platform, or payment service provider (PSP). These modalities are aligned with real-world checkout flows and user expectations, including compatibility with EMV 3-D Secure for online card transactions and SEPA Instant Credit Transfer (SCT Inst) flows for bank transfers. Communication between PSP/bank or payment terminal and Wallet Unit is conducted via standard HTTPS endpoints and OAuth 2.0-based authorisation protocols, ensuring compatibility with existing financial APIs and merchant platforms.

In the future, it is planned to support proximity (“contactless”) scenarios with use of ISO/IEC 18013 standards as well.

Data model, format and proof mechanisms

The Payment Authentication process leverages a Verifiable Credential that proves the user’s identity (if required) and payment credentials and, where applicable, additional attributes such as age, license status, or residence. These credentials are in a form of a verifiable attestations, formatted as SD-JWT Verifiable Credentials or other in the future.

In the context of payments, attestations are often presented to fulfil SCA requirements defined under PSD2, with wallet-based authentication serving as the “possession” and biometric/PIN acting as the “inherence” or “knowledge” factor. Selective disclosure is supported, allowing users to share only the required attributes for a specific payment context, without revealing full identity unnecessarily.

Each of the flows may provide with its own “schema” for transaction processing. Each schema contains (at least) the following data structures:

1. transactional data schema (for constructing a presentation request),
2. display schema (for rendering information displayed to the Wallet User),
3. localisation information (for proper translations of information for the Wallet User),
4. response schema (for delivering an authentication code as proof of consent for the payment transaction).

SCA Attestation is an attestation issued by a Payment Service Provider (PSP) into a Wallet Unit that serves as a verifiable digital proof that the Wallet Unit is associated with a specific Payer and their account managed by the PSP. In the payments ecosystem, payment data structures differ between payment methods. Therefore, there will be multiple types and versions of SCA Attestations. The structures of the SCA Attestations applicable to this specification are to be specified in related SCA Attestation Rulebook(s).

Issuance

Attestations in general are issued by trust service providers or governmental entities. In case of payments, related attestations can be issued directly by the Payment Service Provider (such as Banks, acting as a trust service provider) or another entity acting on behalf of this PSP (another trust service provider). These issuers use the OID4VCI protocol to deliver “SCA Attestations” directly to the user’s wallet after a secure User’s identity and Wallet’s validity verifications.

Verification

When a user initiates a payment or responds to a payment request, the EUDI Wallet uses the OID4VP (or ISO/IEC 18013-5 in the future) protocol to present the relevant credential to the relying party (e.g. bank, acquirer, or merchant).

The solution addresses the regulatory requirement that the Payer can give consent to execute a payment directly towards their Payment Service Provider (PSP) in the setup referred to as “Issuer-requested flow”, or via a third party (Payee or Payment Initiation Service Provider/Account Information Service Provider) in the setup referred to as “Third-party-requested flow”.

Trust framework

The Payment Authentication use case relies on the eIDAS 2.0 trust framework to establish legal and technical validity for both identity credentials and authentication signals. All issuers of PID or EAA used in payments must be listed on a Trusted List and conform to the applicable regulatory requirements. Verifiers (such as banks or merchants) resolve trust by validating the issuer’s metadata and cryptographic keys against this federated trust list.

